

# Zach Zukowski

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## AI GOVERNANCE · CONTROL-LAYER ACCOUNTABILITY · DILIGENCE FRAMEWORKS

- 22 MCP + API connections · ~75 DATs analyzed · Diligence-doc review 4hr to 1hr · 8 SSRN papers · 13 federal comment letters
- Patent-pending Canonical-State Architecture for multi-agent AI research (eight USPTO provisional applications, 2026)

### SUMMARY

AI-augmented research and governance-infrastructure builder who turns control-layer and operator-accountability research into machine-readable diligence systems, now extending that accountability lens from stablecoin gateways to AI deployment. Founder of Tokenization Systems; first hire at Borderless Capital, an RIA.

Focus: research-infrastructure engineering, control-layer accountability, machine-readable diligence, agentic workflow automation · MCP server design, Monte Carlo, difference-in-differences, coalition-power analysis · Python, R, DuneSQL.

### EXPERIENCE

#### Founder · Tokenization Systems

*Feb 2026 to Present*

- Authored an eight-paper SSRN-live research program covering the [Control Layer Intensity Index](#) (CLII) for gateway concentration, the [Minimum Viable Equivalence Pack](#) (MVEP) for institutional tokenization diligence, programmable monetary-policy design ([Adaptive Tokenomics](#)), [governance concentration](#), and [operational risk in token economies](#).
- Contributed thirteen comment letters across a federal regulatory program (OCC bank stablecoin engagement [RIN 1557-AF41](#) plus [supplemental control-layer extension](#); [Treasury GENIUS Act state similarity](#); FDIC payment-stablecoin reserve standards [RIN 3064-AG19](#) and stablecoin-subsidary issuance approvals [RIN 3064-AG20](#); FinCEN AML/CFT [FINCEN-2026-0034](#); FinCEN-OFAC PPSI sanctions [FINCEN-2026-0100](#); joint banking AML/CFT [91 FR 18304](#); [Routing the Dollar](#) submitted to the Federal Reserve Fifth Conference on the Dollar's International Roles (Jun 2026); [Governance Concentration](#) in peer review at Frontiers in Blockchain.
- Working dialogue with capital-markets infrastructure operators on stablecoin, deposit-token, tokenized-deposit, and tokenized-equity deployment.

#### Senior Investment Analyst, Digital Assets · Borderless Capital (RIA; first hire)

*Dec 2019 to Feb 2026*

##### Research Function & Automation

- Built automated SEC-filing and press-release monitoring for Digital Asset Treasury Stocks (DATs); analyzed ~75 tokenization offerings in 2025 and invested in 18; AI-assisted diligence workflows reducing subscription-document review from ~4 hours to ~1 hour per deal.
- Built the firm's research function from scratch; led a four-analyst team and personally produced ~25 of its ~125 long-form publications annually; taught 100 teams across nine accelerator programs (2021 to 2025).

##### Governance Standards Advisory

- Translated regulatory change (GENIUS, CLARITY, Singapore MAS, MiCA, state money-transmitter and trust-charter regimes) into product, reserve, routing, and compliance decisions across U.S., Singapore, Cayman / BVI / Bermuda, and EU; coordinated cross-functional partner work spanning Product / Engineering, Risk (operational-risk taxonomy), and Legal.
- Advised 100 portfolio companies 1-on-1 on token mechanism design (stablecoin reserves, governance-token allocation and incentive design, yield wrappers, burn/emission dynamics); standards defined in [Introduction to Tokenomics](#).
- Built quantitative token-economic models forecasting emissions, fee capture, pricing and subsidy-to-revenue trajectories, and governance concentration under wide scenario ranges; supply-equilibrium and slashing dynamics supporting Layer 1 business-model and mechanism-design advisory.

##### Investment & Conviction Calls

- Sourced the **LayerZero seed round** investment for Borderless, returning ~80x; the underlying cross-chain interoperability thesis preceded the firm's Cross-Chain Ecosystem Fund by two years.
- Predicted Thorchain's **\$200M insolvency 13 months in advance**; Borderless held no exposure when the predicted failure mode materialized.
- Shipped 250 portfolio investments (~25% deployment rate) across ~10 institutional funds (~\$500M AUM) from a 1,000+ deal pre-seed-to-Series-C funnel, with build/partner/hold recommendations to committee.

## SELECTED RESEARCH

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- **Canonical-State Architecture.** Filed eight US provisional patent applications in 2026: a five-element coordination system for concurrent AI-agent sessions on a shared version-controlled repository, integrating per-session staging-index isolation, a multi-mode verification engine, a commit-time canonical-state coherence gate, recursive self-validation of discipline codifications, and an operator-confirmed promotion gate; validated across ~2,170 commits and 4 to 6 parallel AI coding sessions over ~4 months, with a companion 73-page methodology preprint. The applications extend the architecture to role-routed cross-session coordination, actual-touched-file collision detection, and a self-describing verification registry, plus separate applications for evidence-grounding of AI-generated assessments and cited values.
- **[Auditing Governance Concentration Beyond Token Allocation.](#)** In peer review at Frontiers in Blockchain. Live-governance audit of 52 token protocols: launch allocation does not predict governance concentration; DePIN governance is more concentrated than DeFi after correction (Cohen's  $d = 0.65$ , a directionally robust medium effect).
- **[Operational Risk in Token Economies.](#)** Operational-risk taxonomy (74 risk vectors, 64 testable controls, 31 program objectives) mapped to a 624-scenario battery; Lido V3 staking-vault case study.
- **[Adaptive Tokenomics.](#)** Industrial control engineering (PID controller) applied to token-supply emission as programmable monetary-policy mechanism; 240-scenario Monte Carlo simulation; cuts worst-case supply volatility ~6x under competitor-shock attack.
- **[Who Burns the Tokens?](#)** Burn-mechanism design (carrier contracts vs. subscriptions) drives demand concentration: Helium's carrier-contract model concentrates ~90% of token burns in 5 purchasers (HHI 0.22) despite ~1M hotspots; coalition-power analysis extends standard concentration measures for single-address risk.
- **[Routing the Dollar.](#)** Submitted to Federal Reserve / FRBNY Fifth Conference on the Dollar's International Roles (Jun 2026). Gateway concentration; CLII across 19 gateways including Visa Direct-adjacent paths.
- **[Minimum Viable Equivalence Pack.](#)** Ten-category institutional diligence standard; finds bank-chartered stablecoin issuers face ~22x the regulatory capital requirements of money-transmitter-chartered issuers at \$40B scale.
- **[Three Tokenization Strategies.](#)** Each of three strategies satisfies different MVEP categories with measurably different strength (Regulated Collateral Circuit, Multi-Issuer Yield Stack, Tokenized Equity Rails); Securitize as cross-strategy concentration risk.
- **[Seven Dollars.](#)** Cross-product framework for major dollar-denominated digital products; CLII / MVEP decoupling; stress propagation through dependency chains (SVB, Circle pool, Paxos).

## TECHNICAL

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**Methods:** policy-shock impact estimation (difference-in-differences), Monte Carlo simulation, PID controller design (industrial control engineering), coalition-power analysis, supply equilibrium modeling, build-versus-partner framework design

**Data:** Dune Analytics, Nansen Pro, Artemis, DefiLlama, CoinGecko, Helius (Solana), Etherscan, Blockscout (EVM), Spacescope, Token Terminal, Tally (DAO governance), rated.network, Beaconcha.in (validator data), FRED (macro); automated SEC filing and press release monitoring

**Stack:** Python (pandas, statsmodels, scipy, matplotlib), R (fixest, did, synthdid), SQL (DuneSQL, Trino), JavaScript / TypeScript; 22 MCP and API connections wrapping the data sources above plus GitHub; Claude.ai + Claude Code + Cursor with 36-skill library and 17-file living-knowledge architecture; n8n, Apify, Git / GitHub

## RECOGNITION

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- Uniswap Foundation governance committee (2025)
- ETHDenver pitch judge (2025)
- **[Introduction to Tokenomics](#)** article publicly endorsed by Binance founder CZ on day of publication

## EDUCATION

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**High Point University** | B.S.B.A., Business Administration (Magna Cum Laude) · Investment Club President  
Lingnan University, Hong Kong (Exchange)

2018